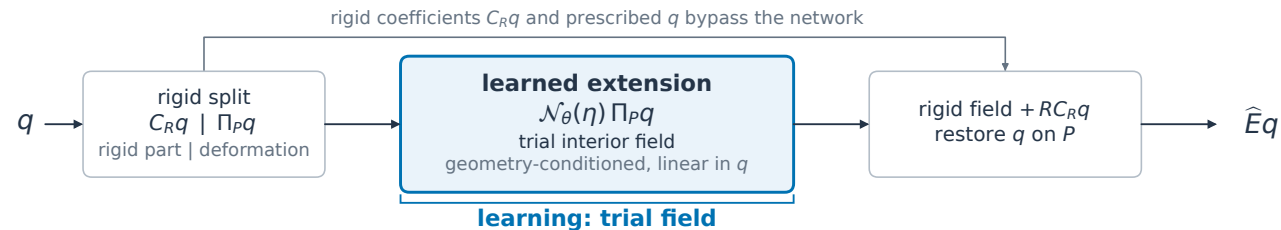


(a) Condensed operator of one cell



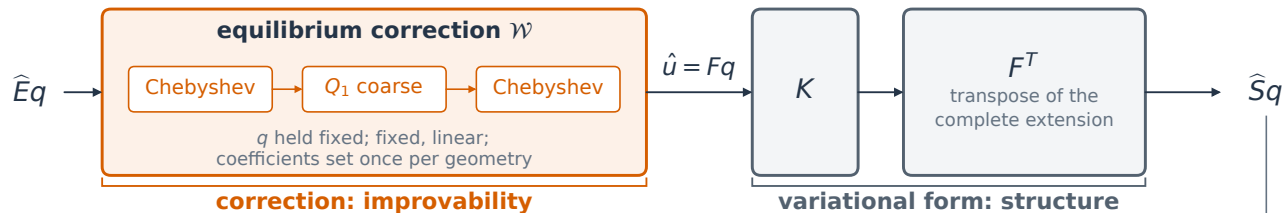
by construction
(Eq. 14)

$J_P \hat{E} = I_P$
admissible

$\hat{E} R_P = R$
rigid-body motion reproduced

$\hat{E}$ linear in q
at fixed geometry

one θ
shared by all cells



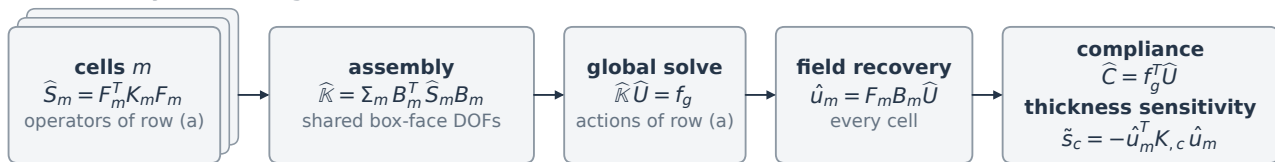
condensed operator

$\hat{S} = F^T K F$
symmetric, ≥ 0 , rigid-body kernel

$\hat{S} - S = H^T A H \geq 0$
error quadratic in the field error exact or shifted coarse solve, spectrum in $(0, b)$

$S \leq \hat{S} \leq \hat{S}_{\text{net}}$

(b) Assembly and design



design iteration: update of the corner parameters τ_c (new geometry η , same network parameters θ ; correction coefficients recomputed)